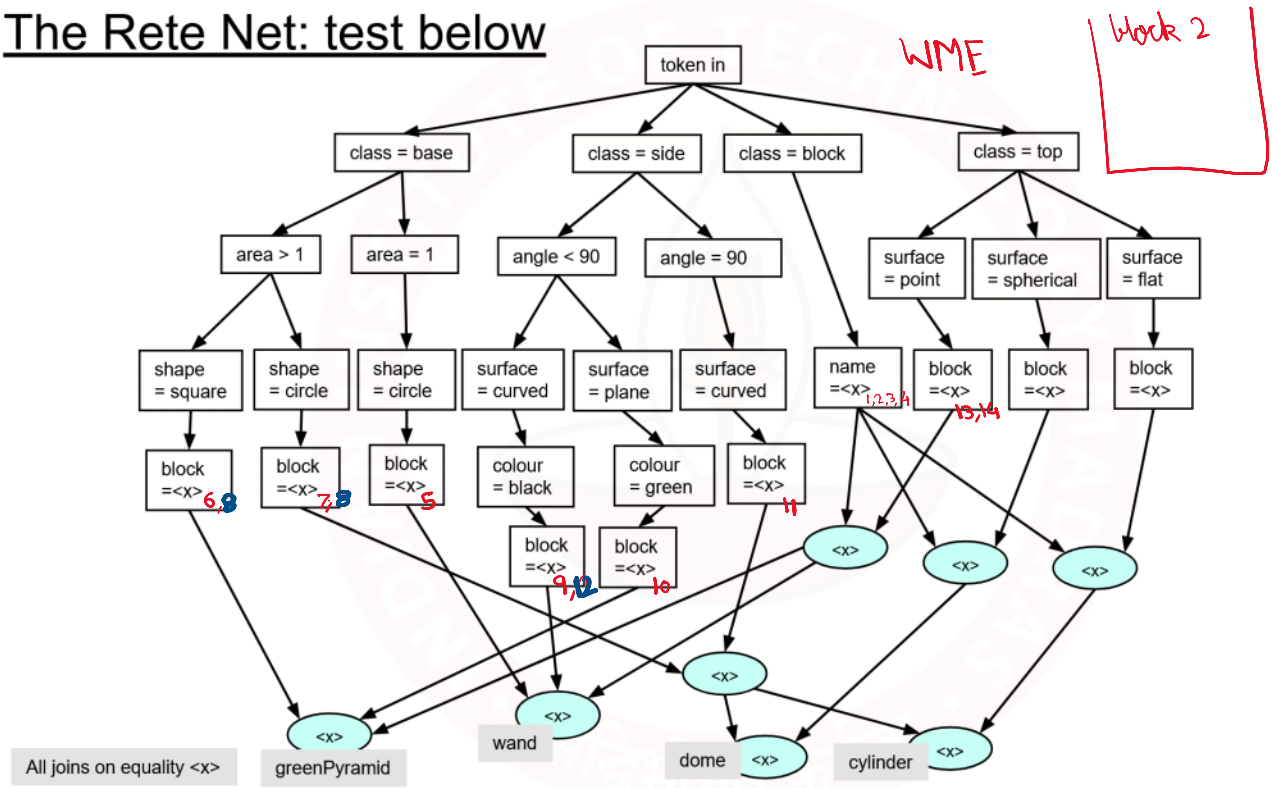
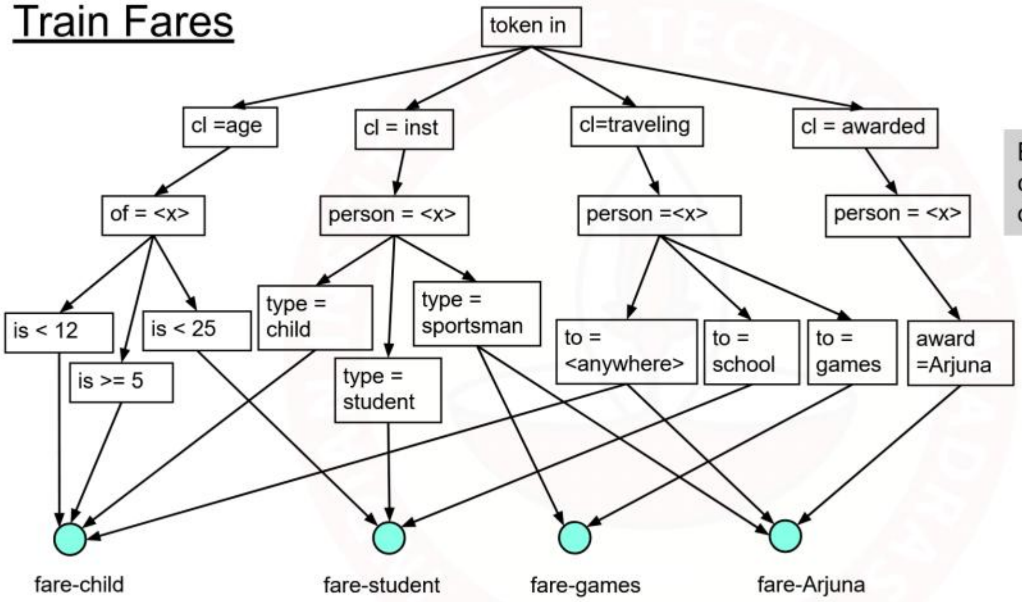
The Rete Net: test below



Train Fares



Exercise: Write the corresponding rules to determine fare category

All Joins on equality of variable <x>

(p fare-child

(person ^name <x>)

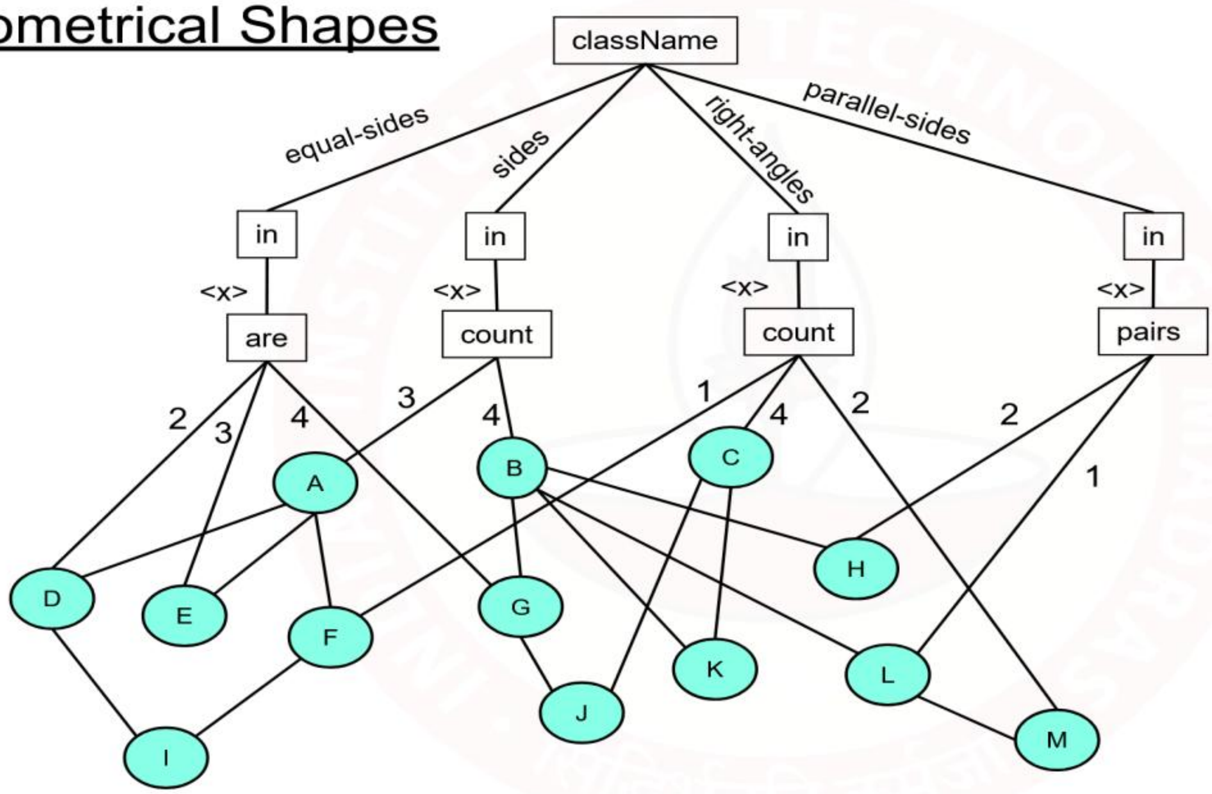
(age ^of <x> ^is {<a> ≥5 <12})

(inst ^person <x> ^type <child>)

(travelling ^person<x> ^to ?anywhere)

```
[illegible]
    (travelling ^person<x> ^to ?anywhere)
-->
    ( ____________ ))

(p fare-student
    (person ^name <x>)
    (age ^of x ^is {<a> >12 <25})
    (inst ^person <x> ^type <student>)
    (travelling ^person<x> ^to <school>)
-->
    ( ______________________ ))

(p fare-games
    (person ^name <x>)
    (inst ^person <x> ^type <sportsman>)
    (travelling ^person<x> ^to <games>)
-->
    ( ______________________ ))

(p fare-games
    (person ^name <x>)
    (inst ^person <x> ^type <sportsman>)
    (travelling ^person<x> ^to ?anywhere)
    (awarded ^award <Arjuna>)
-->
    ( ______________________ ))
```

Geometrical Shapes



A - Triangle

D - Isoceles

E - Equilateral

F - Right-Angled

I - Isoceles Right-Angled

B - Quadrilateral

G - Rhombus

J - Square

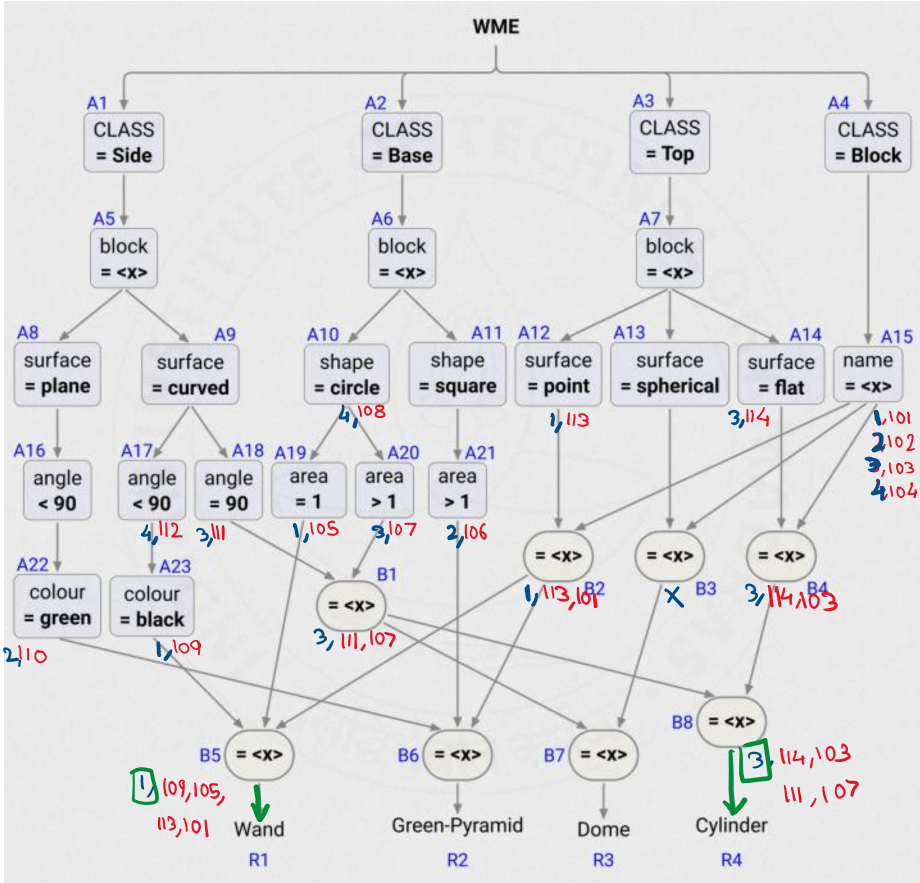
K - Rectangle

C - Any shape with 4 ⊥

H - Parallelogram

L - Trapezoid

M - Right-Angled Trapezoid




```
101. (block ^name 1 ^owner Harry)
102. (block ^name 2 ^owner Hermione)
103. (block ^name 3 ^owner Ron)
104. (block ^name 4 ^owner Draco)
105. (base ^block 1 ^shape circle ^area 1)
106. (base ^block 2 ^shape square ^area 20)
107. (base ^block 3 ^shape circle ^area 10)
108. (base ^block 4 ^shape circle)
109. (side ^block 1 ^angle 80 ^surface curved ^color black)
110. (side ^block 2 ^angle 60 ^surface plane ^color green)
111. (side ^block 3 ^angle 90 ^surface curved)
112. (side ^block 4 ^angle 50 ^surface curved)
113. (top ^block 1 ^surface point)
114. (top ^block 3 ^surface flat)
```

Nodes fire only if all incoming are honored.

Conflict Set = { < R1 101 105 109 113 >, < R4 103 107 111 114 > }

R1: 2 4 5 3 = 14

R4: 2 4 4 3 = 13

A4, A15

A2, A6, A10, A19

```
(p copy-rule
    (rec ^a = 0 ^b <y>)
    -->
    (MAKE (rec ^c <y>)) )
(p swap-rule
    (rec ^a <x> ^b <y> < <x>)
    -->
    (MAKE (rec ^a <y> ^b <x>)) )
(p reduce-rule
    (rec ^a <x> > 0 ^b <y> >= <x>)
    -->
    (MAKE (rec ^a <x> ^b (<y> - <x>))) )

Working Memory
101. (rec ^a 15 ^b 20)
```

102. (rec ^a 15 ^b 5)
103. (rec ^a 5 ^b 15)
104. (rec ^a 5 ^b 10)
105. (rec ^a 5 ^b 5)
106. (rec ^a 5 ^b 0)
107. (rec ^a 0 ^b 5)
108. (rec ^c 5)

GA - Group 2

(7) reduce-rule will fire

MAKE (rec ^a 15 ^b (20-15))

So (rec ^a 15 ^b 5) is created

(8) swap-rule will be satisfied by 102 as $5 < 15$.

MAKE (rec ^a 5 ^b 15)

(9) Consider WMEs are removed after EXECUTE

1) reduce-rule for 103 ($5 > 0$ & $15 \geq 5$)
   MAKE (rec ^a 5 ^b 10)

2) reduce-rule for 104 ($5 > 0$ & $10 \geq 5$)
   MAKE (rec ^a 5 ^b 5)

3) reduce-rule for 105 ($5 > 0$ & $5 \geq 5$)
   MAKE (rec ^a 5 ^b 0)

4) swap - rule for 106 ($0 < 5$)
   MAKE (rec ^a 0 ^b 5)

5) copy - rule for 107 ($0 = 0$)
   MAKE (rec ^c 5)

6) Halt! Because 108 doesn't match

reduce - rule executes 4 times

reduce - rule executes 4 times

Swap-rule fires 2 times

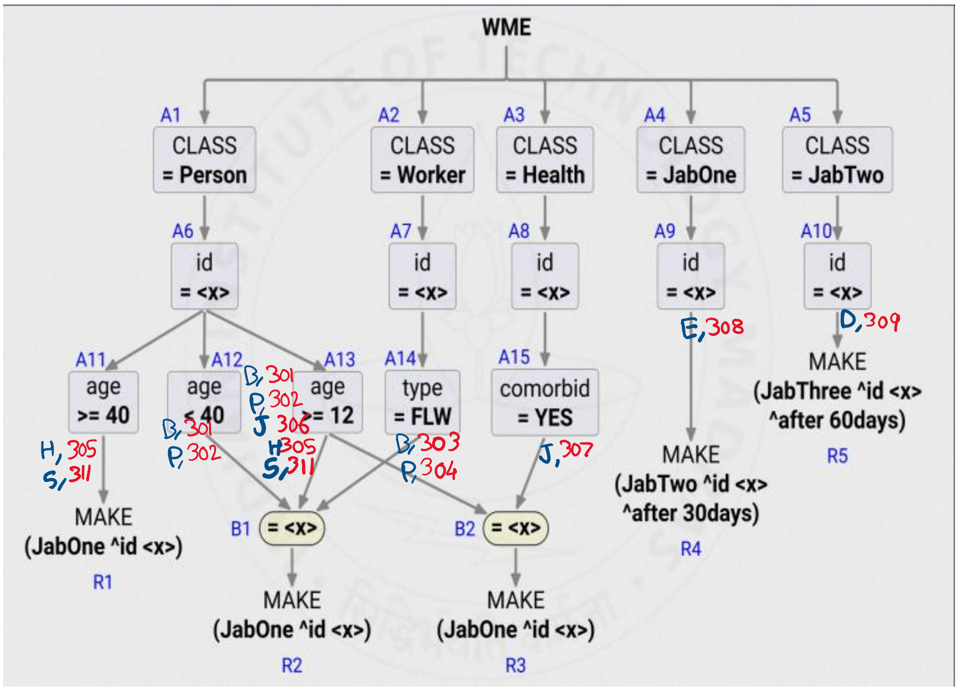
```
(p rule-1
    (rec ^index <a> ^value <x>)
    (rec ^index <b> > <a> ^value <y> < <x>)
    --> ...RHS...
)
(p rule-2
    (rec ^value <x>)
    -(rec ^value <y> < <x>)
    --> ...RHS...
)

Working Memory
201. (rec ^index 3 ^value 17)
202. (rec ^index 1 ^value 20)
203. (rec ^index 2 ^value 23)
```

GA10 GROUP-3

rank

1. rule-1 — 202 201
203 201

rule-2 — 201



~~301.~~ (Person ^id Ben ^age 25)
~~302.~~ (Person ^id Pat ^age 20)
~~303.~~ (Worker ^id Ben ^type FLW)
~~304.~~ (Worker ^id Pat ^type FLW)
~~305.~~ (Person ^id Hue ^age 53)
~~306.~~ (Person ^id Joe ^age 18)
~~307.~~ (Health ^id Joe ^comorbid YES)
~~308.~~ (JabOne ^id Emy)
~~309.~~ (JabTwo ^id Don ^after 30days)
~~310.~~ (JabThree ^id Ann ^after 60days)
~~311.~~ (Person ^id Sam ^age 86)

CS = { <R1 305> (3), <R1 311> (3), <R2 301 303> (4 + 3 = 7), <R2 302 304> (4 + 3 = 7),

<R3 306 307> (3 + 3 = 6), <R4 308> (2), <R5 309> (2) }